

KATHMANDU UNIVERSITY
End Semester Examination
March/April, 2025

Marks Scored:

Level : B.E./BIT
Year : I

Course : MATH 101
Semester : I

Exam Roll No. :

Time: 30 mins.

F. M. : 20

Registration No.:

Date : 02 APR 2025

SECTION "A"
[10 Q. $\times$ 1 = 10 marks]

Fill in the blank space(s) by writing the most appropriate word(s) or symbol(s).

1. The domain of f is A and domain of g is B then the domain of fg is _____.
2. $\lim_{x \rightarrow \frac{5}{2}} \left[\frac{x}{2} \right] =$ _____ where $[]$ denote it's usual meaning
3. The function $y = |x|$ is differentiable at every point of x except _____.
4. The column of a matrix A are linearly independent if and only if the echelon form of A has _____.
5. The vertical asymptotes of the function $y = \frac{2x}{x-3}$ is _____.
6. The critical point of $y = x^2 - 2x$ is _____.
7. If the series is convergent and whose sum of the n -th term is $y = \frac{2n}{3n+5}$, then the limit is _____.
8. If a matrix A has n columns, then $\text{rank } A + \dim(\text{Null } A) =$ _____.
9. The scalar 7 is an eigenvalue of A , if and only if the equation $Ax =$ _____.
10. If a set contains more vectors than there are entries in each vector, then the set is _____.

SECTION "B"

[10 Q. × 1 = 10 marks]

Fill in the blank space(s), **DO NOT TICK**, by selecting the most appropriate answers from among the given ones.

11. If $f(x) = -\sqrt{x}$ shifted right by 3 units, then the new function is given by _____.
 $[-\sqrt{x-3}, \quad \sqrt{x-3}, \quad \sqrt{x+3}, \quad -\sqrt{x+3}]$
12. If f' changes from positive to negative at c , then f has: _____ at c
 $[\text{Local maxima}, \quad \text{Local minima}, \quad \text{Absolute minima}, \quad \text{No extreme values}]$
13. The interval in which function $f(x) = x^2 - 4x + 5$ is increasing is _____.
 $[(0, \infty), \quad (2, \infty), \quad (-\infty, 2), \quad (4, \infty)]$
14. At $x=1$, the graph of the function $y = 4x^2 - 2x$ is _____.
 $[\text{Concave up}, \quad \text{Concave down}, \quad \text{Stationary}, \quad \text{Decreasing}]$
15. Which of the following is the same as the definite integral $\int_a^b f(x) dx$: _____.
 $[\int_a^b f(x) da, \quad \int_a^b f(x) db, \quad -\int_b^a f(x) dx, \quad -\int_a^b f(x) dx]$
16. The improper integral $\int_0^\infty \frac{dx}{x^2+4}$: _____.
 $[\text{Diverges to infinity}, \quad \text{Converges to zero}, \quad \text{Converges to } \frac{\pi}{4}, \quad \text{Converges to } \frac{\pi}{2}]$
17. The series $\sum n^{-p}$ is convergent iff _____.
 $[p < 1, \quad p > 1, \quad p = 1, \quad p \leq 1]$
18. If one row in an echelon form of an augmented matrix is $[0 \quad 0 \quad 0 \quad 5]$ then the associated linear system has _____.
 $[\text{exactly one solution}, \quad \text{two solution}, \quad \text{infinitely many solution}, \quad \text{no solution}]$
19. The maximum rank of a 3×3 matrix is: _____.
 $[0, \quad 1, \quad 2, \quad 3]$
20. The scalar λ is a characteristic root of the matrix A if _____ is singular
 $[A, \quad \lambda I, \quad \lambda - AI, \quad A - \lambda I]$

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02 APR 2025

SECTION "C"

[3 Q. × 7 = 21 marks]

1. Define continuity of a function $f(x)$ at a point $x=a$. Discuss different types of discontinuity with examples. Prove that differentiable functions are continuous at the point but converse need not be true. [1+3+3=7]
2. State the first and second fundamental theorem of calculus. Find the area of the surface generated by revolving the curve $y = 2\sqrt{x}$, $1 \leq x \leq 2$ about the x -axis. [2+5=7]

OR

Define Roll's theorem and Mean value theorem. Show that the equation $x^3+3x+1 = 0$ has exactly one real solution. Find the value(s) of c that satisfies the equation

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

for the function $f(x) = x^3 - x^2$ in the interval $[-1, 2]$.

[2+2+3=7]

3. Define column space of a matrix. Find the null space and column space of a matrix A , given by:

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Also, find their dimensions.

[2+3+2=7]

SECTION "D"

[6 Q. × 4 = 24 marks]

4. Find $\frac{dy}{dx}$ (**ANY TWO**):

a. $y = \sqrt{\cos(2x+1)}$

b. $y^2 - x^2 = \sin(xy)$

c. $x^m y^n = (x+y)^{m+n}$

5. Evaluate (**ANY TWO**):

(a) $\int e^x \cos(x) dx$

(b) $\int \frac{e^{\tan^{-1}(x)}}{1+x^2} dx$

(c) $\int_2^3 \frac{x^2}{x-1} dx$

P.T.O.

6. State ratio test for the convergence of an infinite series and Test the convergence of the series: $\frac{2}{3!} - \frac{2^2}{5!} + \frac{2^3}{7!} - \frac{2^4}{9!} + \dots$

7. Find the volume of the solid of revolution of curve $y = \sqrt{x}$, and the line $y = 1, x = 4$, about the line $y = 1$.

OR

Find the length of the curve $y = x^{3/2}$, from $x = 0$ to $x = 4$.

8. For what value of h will y be in span of $\{v_1, v_2, v_3\}$. If:

$$v_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, \quad v_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \quad y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$

9. Find eigenvalues and eigenvectors of matrix:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

SECTION "E"

[5 Q. $\times$ 2 = 10 marks]

10. Find all asymptotes of $y = \frac{x^2-1}{x^2+1}$.

11. Evaluate $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1}$.

12. Find the domain of $y = y = -\sqrt{x}$

13. Show that the transformation $T : R^2 \rightarrow R^2$ defined by $T(x, y) = (x - y, x + y)$ is a linear transformation.

14. Let

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Then determine whether $\{v_1, v_2, v_3\}$ is linearly dependent or independent.